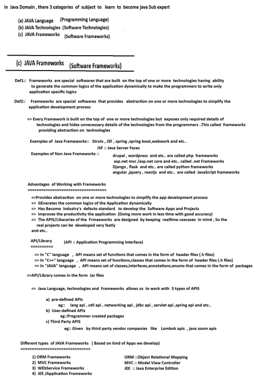

In Java Domain , there 3 categories of subject to learn to become java Sub expert

(a) JAVA Language (Programming Language)
(b) JAVA Technologies (Software Technologies)
(c) JAVA Frameworks (Software Frameworks)

(c) JAVA Frameworks (Software Frameworks)

Def1:: Frameworks are special softwares that are built on the top of one or more technologies having ability to generate the common logics of the application dynamically to make the programmers to write only application specific logics

Def2:: Frameworks are special softwares that provides abstraction on one or more technologies to simplify the application development process

=> Every Framework is built on the top of one or more technologies but exposes only required details of technologies and hides unnecessary details of the technologies from the programmers ..This called frameworks providing abstraction on technologies

Examples of Java Frameworks:: Struts , JSF , spring ,spring boot,webwork and etc..
JSF :: Java Server Faces

Examples of Non Java Frameworks :: drupal , wordpress and etc.. are called php frameworks
asp.net mvc /asp.net core and etc.. called .net Frameworks
Django , flask and etc.. are called python frameworks
angular ,jquery , reactjs and etc.. are called JavaScript frameworks

Advantages of Working with Frameworks
====================================

=>Provides abstraction on one or more technologies to simplify the app development process
=> GEnerates the common logics of the Application dynamically
=> Has Become industry's defacto standard to develop the Software Apps and Projects
=> Improves the productivity the application (Doing more work in less time with good accuracy)
=> The APIS/Libararies of the Frmaeworks are designed by keeping realtime usecases in mind , So the real projects can be developed very fastly
and etc..

API/Library (API :: Application Programming Interface)
==========

=> In "C" language , API means set of functions that comes in the form of header files (.h files)
=> In "C++" language , API means set of functions,classes that comes in the form of header files (.h files)
=> In "JAVA" language , API means set of classes,interfaces,annotations,enums that comes in the form of packages

=>API/Library comes in the form Jar files

=> Java Language, technologies and Frameworks allows us to work with 3 types of APIS

a) pre-defined APIs
    eg:: lang api , util api , networking api , jdbc api , servlet api ,spring api and etc..
b) User-defined APIs
    eg::Programmer created packages
c) Third Party APIS
    eg:: Given by third party vendor companies like Lombok apis , java zoom apis

Different types of JAVA Frameworks (Based on kind of Apps we develop)
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1) ORM Frameworks
2) MVC Frameworks
3) WEbService Frameworks
4) JEE /Application Frameworks

ORM ::Object Relational Mapping
MVC :: Model View Controller
JEE :: Java Enterprise Edition